

Cyber Crime: Understanding Threats in the Digital Age

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Outline

- 1 Introduction & What is Cyber Crime?
- 2 Major Types of Cyber Crimes
- 3 How Cyber Criminals Operate
- 4 Impacts & Statistics
- 5 Prevention & Best Practices
- 6 Legal Framework
- 7 Conclusion

What is Cyber Crime?

- Criminal activity using computer, network, or digital device as tool, target, or venue
- No borders: attacker in one country → victim in India in seconds
- Asymmetric: small group can cause billions in damage

Quick History

- 1980s–90s: Early viruses, hacking for fun
- 2000: India's IT Act 2000
- 2010s: Ransomware & state-sponsored attacks rise
- 2025–2026: AI-powered attacks, supply-chain breaches dominate

Why Care in India? (2025–2026 Reality)

24% rise in reported cases in 2025

- **28.15 lakh** cases reported (up from 22.68 lakh in 2024)
- Financial loss: **22,495 crore**
- Investment scams: 76% of total losses
- Massive malware detections: 369 million+ across millions of endpoints (Seqrite/DSCI 2025 report)

Source: Ministry of Home Affairs, DSCI India Cyber Threat Report 2025

Major Types of Cyber Crimes

- ❶ **Financial Crimes** (most common in India)
 - Phishing, vishing, smishing
 - Investment scams & digital arrest
 - Business Email Compromise (BEC)
- ❷ **Data Breaches & Identity Theft**

- ❸ **Malware & Ransomware**
 - Locks files, demands payment
 - Still #1 organisational risk
- ❹ **Cyber Terrorism & Infrastructure Attacks**
- ❺ **Emerging 2025–2026**
 - Deepfake fraud (AI voice/video)
 - Supply-chain attacks
 - AI as weapon & target

Attack Lifecycle & Delivery Methods

Typical cycle: Recon → Weaponisation → Delivery → Exploitation → Installation → C2

Top Delivery Channels (2025–2026)

- Email (82% of malware delivery)
- Malicious websites & phishing kits
- SMS, fake apps, USB drives
- AI-generated convincing content

Game-changers

- AI-enhanced phishing & deepfakes
- Data extortion (no encryption needed)
- Multichannel attacks (email + call + SMS)

Cyber crime = third-largest “economy” if a country!

- Global annual cost: **\$10.5 trillion** (widely cited 2025 projection)
- Average data breach cost: ~\$4.88 million
- Ransomware damages: tens of billions projected

India 2025

- 28.15 lakh cases
- 22,495 crore lost
- Weekly attacks: ~3,195 per organisation (Check Point)
- Education & healthcare heavily targeted

- **Digital Arrest Scams (India):** Fraudsters pose as CBI/ED via video calls → crores lost
- **Change Healthcare (USA, 2024–25 spillover):** Ransomware disrupted prescriptions for weeks
- **Coupang (South Korea, 2025):** Insider breach exposed 34 million users
- **Supply-chain trend:** One weak vendor → thousands infected

Lesson: Human element + supply chain = biggest vulnerabilities

Prevention: Individual Level (5 Key Steps)

- ① Strong, unique passwords + password manager
- ② Enable 2FA/MFA everywhere possible
- ③ Never click suspicious links — type URL manually
- ④ Keep all software & OS updated (patches close 90% vulnerabilities)
- ⑤ Be sceptical of urgent money/data requests

95% of breaches are preventable with basic hygiene!

Prevention: Organisational Level

- Zero-trust architecture
- Mandatory MFA + regular training (human error = #1 cause)
- Offline backups + tested incident response plan
- Cyber insurance (read fine print!)
- Monitor supply chain vendors

Tip: Run phishing simulations regularly

India – Key Laws

- IT Act 2000 (amended)
 - Sec 66: Hacking → up to 3 yrs + 5 lakh
 - Sec 66C: Identity theft → up to 3 yrs + 1 lakh
 - Sec 66F: Cyber terrorism → life imprisonment
- Digital Personal Data Protection Act (DPDPA) 2023

Global: UN Convention against Cybercrime (2025+ signatories)

Report: cybercrime.gov.in | 1930 helpline | local police + CERT-In

Think before you click. Update everything. Report fast.

Final Messages

- Cyber crime affects **everyone** — not “someone else’s problem”
- Awareness + habits + laws = powerful defense

Thank you! Questions?

Overview: Iran-US Conflict

- Joint US-Israel strikes (Operation Epic Fury / Roaring Lion) began Feb 28, 2026.
- Killed Supreme Leader Khamenei & senior officials; targeted nuclear/missile sites.
- Cyber played major role: Pre-strike disruptions + post-strike retaliation threats.
- Iran: Near-total internet blackout (connectivity 4-54% normal); limits state-sponsored ops.
- Focus: US/Israel offensive cyber success vs. Iranian/pro-Iran low-level threats.

US/Israel Offensive Cyber Operations

- US Cyber Command + Space Command: Disrupted Iranian comms, sensors, defenses pre-strike.
- “Non-kinetic” effects blinded coordination (confirmed by Gen. Dan Caine).
- Hacks included:
 - BadeSaba prayer app (5M+ downloads): Pushed anti-regime messages urging defection/amnesties.
 - Defacements of IRNA/news sites, IRGC systems, government services.
 - Reports of hijacked traffic cams, TVs for psyops/surveillance.
- Contributed to nationwide digital blackout + regime restrictions.

Iranian/Pro-Iran Cyber Retaliation (Observed)

- Limited major state-sponsored attacks so far (internet blackout hampers IRGC/APT groups like APT35/APT42).
- Surge in hacktivist activity (60+ groups claimed; pro-Iran, pro-Russian proxies):
 - DDoS, defacements, data leaks on US/Israel/GCC targets (e.g., banks, ports, healthcare).
 - Claims (often overstated) of disruptions in Middle East, US, Asia.
- Examples: DieNet DDoS on US port; FaD Team data exfil from township.
- Phishing/SMS campaigns (e.g., RedAlert malware lures).

Threat Assessments & Warnings (March 2026)

- Unit 42 (Palo Alto): Escalated phishing, hacktivist ops, cybercrime; nation-state activity mitigated short-term.
- CrowdStrike/Google: No large-scale Iranian campaigns; expect disruptive attacks on critical infrastructure (energy, finance, water).
- DHS/CISA/FBI: Low-level hacktivist threats likely (DDoS, defacements); exploit vulnerable OT/IoT (e.g., Hikvision cams).
- Banks/finance: High alert for DDoS/phishing.
- Broader: 150+ hacktivist incidents claimed early March; spillover risks global.

Key Differences from Prior Tensions

- Offensive cyber dominant (US/Israel disrupted Iran heavily).
- Iranian response asymmetric/low-intensity via proxies/hacktivists (plausible deniability).
- No confirmed major successful attacks on US critical infrastructure yet.
- Historical patterns (e.g., 2025 spikes) suggest escalation possible if conflict prolongs.

Recommendations & Outlook

- Patch vulnerabilities, monitor phishing/DDoS (CISA/FBI guidance).
- Critical sectors: Strengthen OT defenses, multi-factor auth.
- Vigilance high: Situation fluid; potential for wipers, ransomware, espionage.
- Outlook: Iranian ops may ramp up as connectivity restores; proxies fill gaps.

Sources (Key References)

- Unit 42 Threat Brief (March 2026)
- Reuters, Nextgov, TechCrunch, Politico reports
- CrowdStrike/Google Threat Intel
- DHS Intelligence Assessment
- NetBlocks, Recorded Future, Sophos